A Numerical Bridge: Pion Mass, QCD String Tension, and the Lepton Sector

A Cross-Sector Consistency Estimate — What It Shows and What It Does Not

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Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it.

Abstract

This paper reports a numerical bridge connecting the pion mass, the QCD string tension, and the lepton-sector scale within the rope framework, treating the rope tension as the common thread. The bridge is a consistency estimate — it checks that the scales line up under the rope identification — rather than a derivation of any one quantity from the others. We report the estimate and state its consistency-check status plainly. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants; the lepton-ratio caveat below is pinned in the test suite.

> **Scope of this paper**
>
> CLAIM: a numerical bridge showing the pion mass, QCD string tension, and lepton scale are consistent under the rope-tension identification.
>
> NOT CLAIMED: a derivation of the pion mass or string tension. A cross-sector consistency estimate; status Modeled.

1. Rope Tension as the Common Scale

The rope framework identifies a single tension underlying confinement-scale and lepton-scale physics. If that identification holds, the pion mass, the QCD string tension, and the lepton scale must be mutually consistent — not independent inputs. This paper checks that consistency numerically.

2. The Bridge and Its Status

The scales line up under the rope-tension identification at the level of a consistency estimate. The paper is explicit that this is a check, not a derivation: it shows the rope identification is not contradicted by these scales, which is weaker than deriving them. Reported as a modeled cross-sector consistency result.

> **Status**
>
> Pion / string-tension / lepton-scale consistency — Modeled (cross-sector estimate).

Derivation of any one scale from the others — Open.

Programme credit: the core Rope Hypothesis is due to Bill Gaede. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.